

Abdelrahman Hany *Computer Engineer*

✉ abdelrahman.hany003@gmail.com ☎ +201022333420  Abdelrahman Hany  DopeBiscuit
📍 Cairo, Egypt  Portfolio

EDUCATION

2021 Oct – 2026 Jul **Bachelor of Science in Computer Engineering**
Cairo, Egypt *Ain Shams University - GPA: 3.43*

PROFESSIONAL EXPERIENCE

- 2021 Oct – Present **Part-time Freelance Developer** 
- Executed **100+ freelance automation projects**, transitioning from G Suite scripting to custom Python and **AI agents**.
 - Built **end-to-end** solutions integrating **web scraping**, **custom APIs**, and **cloud services** for complex enterprise workflows.
- 2025 Feb – 2025 Sep **System Architect & Head of Communication** 
Krakow, Poland *ASU ROAR — European Rover Challenge (ERC)*
- Designed a **TP-Link mesh network** and conducted **RF testing** to establish **robust command/video links** for competition missions.
 - Co-developed a **UART-to-CAN bus interfacer** for **reliable, high-speed data flow** between the **NVIDIA Jetson (ROS)** and microcontrollers.
 - Resolved **critical synchronization errors** by restructuring the **ROS TF tree** (transform frames), restoring the localization and mapping pipeline.
 - Configured **Gazebo & RViz** environments using **URDF/Xacro** models to enable high-fidelity **simulation and autonomy testing**.
 - Instructed **100+ recruits** across **two technical workshops** — **ROS TF Trees & Frame Transformations (4.9/5 feedback)** and **Perception Systems**
- 2026 Aug – 2026 Sep **C++ Instructor**
Eduvera Educational Platform
- Developed and taught a **48-hour, 12-session C++ curriculum (fundamentals through OOP)** for Ain Shams University engineering students, including all **labs** and **milestone projects**

PROJECTS

- 2025 Sep – Present **RUNE (Microsoft-Sponsored Graduation Project)** 
Low-Code Automation Platform
- Architected the DSL code generator translating a single JSON schema into **TypeScript, Python, and Go types** across a 4-language monorepo; authored the design RFC
 - Built the **OAuth2 BYO-client flow** (connect, token refresh, Bearer injection) powering all **integration nodes**
 - Engineered the **Go integration node framework** enabling extensible provider/service/tool dispatch, shipping **Gmail** as the first integration
 - Aligned validation logic (**Pydantic + Zod**) across API and frontend to eliminate cross-layer contract **drift**

- Shipped **24** merged PRs, **18** filed issues, and **2** authored RFCs on a 9-person, 5-service team

2025 May	Cartnest E-commerce Platform <i>Full-stack e-commerce site (Team Project)</i> • Co-developed database schema and implemented ORM classes. • Improved performance and scalability via data fragmentation and sharding.
2025 Apr	CHRONOS CPU Scheduler <i>Python desktop app for visualizing CPU scheduling (Team Project)</i> • Built desktop app with FCFS, threading, and signal-driven step simulation. • Added real-time Gantt chart and process control for 4 schedulers.
2024 May	GPS Tracking System <i>Live Tracking using Tiva C Microcontroller (Team Project)</i> • Architected MCU-server communication API with a GUI. • Delivered an EEPROM driver supporting 500+ coordinate storage.

SKILLS & LANGUAGES

Programming Languages — C++, Python, Java, JavaScript (JS), Google Apps Script (GAS), Golang, Verilog, SQL

Robotics & Embedded — ROS, Gazebo, RViz, PCL, OpenCV, CAN Bus, URDF, TF Tree

Frameworks/Libraries — React, NextJs, React Flow, FastAPI, Flask

Tools — Git, Docker, Jupyter, VS Code, Cisco Packet Tracer

Operating Systems — Windows, Linux (Ubuntu)

Spoken Languages — Arabic (Native), English (Fluent), German (Basic)

CERTIFICATIONS

2025 Aug	NTI: AWS Cloud Architect
2024 Sep	MIT 6.86x: From Linear Models to Deep Learning using Python
2024 May	MIT 6.431x: Probability - The Science of Uncertainty and Data
2023 Dec	Cisco Certified Network Associate
2022 Nov	CS50's Introduction to Computer Science

LEADERSHIP & VOLUNTEERING

2023 – 2025	Teaching & Mentorship — IEEE/GDSC <i>Technical Workshop Instructor</i> • Instructed 40+ students in networking fundamentals (TCP/IP, subnetting) for IEEE & GDSC
-------------	---

AWARDS & HONORS

2024 May	ASU AI Engineering Hackathon — 2nd Place Developed a QA AI-powered proof-of-concept for an Ain Shams Competition. Also competed in the Nokia Digital Design Hackathon, NASA Space Apps Challenge, and ECPC'23.
----------	---